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Polymers

Q.1. Define the term monomer, polymer and polymerization.

Ans: A monomer is a small molecule that can be bonded together with identical or similar monomers to form a larger molecule called polymers. Monomers are the basic building blocks of polymers. They are typically simple molecule with two or more reactive sites that can form covalent bonds with other monomers.

A polymer is a large molecule made up of many repeating units of monomers. Polymers can be natural or synthetic. Natural polymers are found in nature, such as proteins, carbohydrates and nucleic acids. Synthetic polymers are made by humans such as plastics, rubber and fibers.

Polymerization is the process of forming a polymer by linking together monomers. There are two main types of polymerization reactions: addition polymerization & condensation polymerization.

Addition polymerization:- In addition polymerization, momers are added to each other in a chain reaction. This type of polymerization is typically used to make synthetic polymers, such as polyethylene & polypropylene.

Condensation Polymerization:- In condensation polymerization, monomers are joined together by a condensation reaction, which involves the loss of a small molecule, such as water or methanol. This type of polymerization is typically used to make natural polymers, such as proteins and polyesters.

Q.2. Define/Discuss classificat^n of polymers.

Ans:- Polymers can be classified in various ways based on their source, structure, polymerization method and molecular forces. Here's an overview of the main classification methods:

Classification Based on Source:-

1. Natural Polymers:- These polymers occur naturally in plants & animals and are typically biodegradable.
   Ex- cellulose, proteins, rubber & scratch

2. Synthetic Polymers:- These polymers are man-made and are not found in nature.
   Ex- polyethylene, polypropylene, polystyrene & nylon.

3. Semi-synthetic Polymers:- These polymers are derived from natural polymers but have been chemically modified.
   Ex- cellulose acetate & cellulose nitrate.

Classification Based On Structure:-

1. Linear Polymers:- These polymers have a linear chain of repeating units.
   Ex- polyethylene

2. Branch-Chain Polymers:- These polymers have a chain of repeating units with side branches.
   Ex- polystyrene & low-density polyethylene

3. Network Polymers:- These polymers have a crosslinked network of repeating units, giving them a three-dimensional structure.
   Ex- rubber & thermosetting plastics.

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Classification Based on Polymerization Method-

1. Addition Polymers:- These polymers are formed by the addition of monomers to a growing chain.
Ex- polyethylene, polypropylene & polyvinyl chloride.

2. Condensation polymers:- These polymers are formed by the condensat" reaction of monomers with the loss of a small molecule like water or methanol.
Ex- Nylon, polyester & polyurethane.

Classification Based on Molecular Forces:-

1. Elastomers:- These polymers have weak intermolecular forces, allowing them to stretch and return to their original shape.
Ex- rubber & buna-S.

2. Fibers:- These polymers have strong intermolecular forces, giving them high tensile strength & making them suitable for textiles.
Ex- nylon-6,6 & polyester.

3. Thermoplastics:- These polymers have intermediate intermolecular forces, allowing them to be softened & reshaped upon heating.
Ex- polyethylene & Polyvinyl chloride.

4. Thermosetting Polymers:- These polymers undergo irreversible chemical changes upon heating, forming a rigid, crosslinked network.
Ex- epoxy resins, bakelite & silicones.

Q.3. Explain forming mechanism of polymerization.
Ans:- Polymerization is the process of forming a polymer by linking

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together of monomers. There are two main types of polymerization reactions: addition polymerization & condensation polymerization.

Addition Polymerization:- Addition polymerization occurs when monomers are added to a growing chain without the elimination of any small molecule. This type of polymerization is typically used to make synthetic polymers, such as polyethylene & polypropylene.

The mechanism of addition polymerization can be summarized in three steps:-

1. Initiation:- An initiator is added to the monomers to create a reactive site or free radical.

2. Propagation:- The reactive sites or free radical attacks a monomer, forming a covalent bond & creating a new reactive site or free radical. This process continues as more monomers are added to the growing chain.

3. Termination:- The polymerization reaction is terminated when two reactive sites or free-radicals combine, or when a reactive site or free radical reacts with another molecule, such as initiator or a monomer.

Condensation Polymerization:- Condensation polymerization occurs when monomers are linked together by a condensation reaction, which involves the loss of a small molecule, such as water or methanol. This type of polymerization is typically used to make natural polymers such as proteins, & polyesters.

The mechanism of condensation polymerization can be summerized in two steps:-

1. Nucleophile attack:- A nucleophilic group on one monomer attacks the electrophilic group on another monomer, forming a covalent bond.

2. Elimination reaction:- A small molecule, such as water or methanol is eliminated from the reaction products. This process continues as more monomers are linked together.

(i) Free-radical mechanism / polymerization:-

Free-radical polymerization is a type of chain-growth polymerizatn in which monomers are linked together by the successive addition of free radicals. Free radicals are molecules that have an unpaired electron, making them highly reactive and able to initiate and propagate polymerization reactions.

Initiation:- The first step in free-radical polymerizatn is initiation, which involves the creation of free radicals from the monomers. This

Propagation:- Once a free radical is formed, it can react with a monomer, adding it to its chain and creating a new free radical. This process continues as more monomers are added to the growing chain. The monomer that reacts with the free radical is typically the one that is most stable with the free radical.

Termination:- The polymerization reaction is terminated when two free radicals combine with each other or when a free radical reacts with another molecule, such as an initiator or a monomer.

Chain Transfer:- Chain transfer reactions can occur during free-radical Polymerization, which can affect the molecular weight & branching of the polymer. In a chain transfer reaction, a free-radical transfers a hydrogen atom or a halogen atome to another molecule, ending the growing chain & starting a new one.

(ii) Cationic Polymerization:-

Ans:- Cationic polymerization is a type of chain-growth polymerizatⁿ in which a cationic initiator transfers charge to a monomer, which then becomes reactive. This reactive monomer goes on to react similarly with other monomers to form a polymers. The types of monomers necessary for cationic polymerizatⁿ are limited to alkenes with electron-donating substituents and heterocycles. Similar to anionic polymerization reactions, cationic Polymerization reactions are very sensitive to the type of solvent used.

Mechanism Of Cationic Polymerization:-

The mechanism of cationic Polymerization can be summarised in three steps.

Initiation:- A cationic initiator is added to the monomers to create a reactive carbocation. Initiators for cationic Polymerization typically include strong lewis acids, such as BF_3 or $AlCl_3$.

Propagation:- The carbocation attacks a monomer, forming a covalent bond and creating a new carbocation. This process continues as more monomers are added to the growing chain.

Termination:- The polymerization reaction is terminated when the carbocation reacts with a nucleophile, such as water, an alcohol, or an amine. This quenches the carbocation and prevents further polymerizat".

(iii) Anionic polymerization:- Anionic polymerization is a type of chain-growth polymerization in which monomers are linked together by the successive addition of carbanions. Carbanions are negatively charged ions that have an unpaired electron on a carbon atom, making them highly reactive & able to initiate & propagate polymerization reactions.

Mechanism of anionic polymerization:-
The mechanism of anionic polymerization can be summarized in three steps:

Initiation:- An initiator is added to the monomers to create a reactive carbanion. Initiators for anionic polymerization typically include strong bases such as alkali metals or alkyllithium reagents.

Propagation:- The carbanion attacks a monomer, forming a covalent bond & creating a new carbanion. This process continues as more monomers are added to the growing chain.

Termination:- The polymerization reaction is terminated when the carbanion reacts with a proton source, such as water, an alcohol, or an acid. This neutralizes the carbanion & prevents further polymerization.

(iv) Zieglar-Natta Polymerization:- Zieglar-Natta polymerization is a type coordination polymerization used to produce polymers of alkenes. It was developed by Karl Zieglar and Giulio Natta in 1950s and it is one of the most important methods for producing polymers today. Zieglar-Natta polymerization involves the use of a transition metal catalyst to initiate & propagate the polymerization reaction. The catalyst is typically a metal hallide, such as $TiCl_4$ or $ZrCl_4$, & it is activated by an organoaluminium compound, such as ~~t~~ ($Al(C_2H_5)_3$). The activated catalyst then reacts with an alkene monomer, such as ethylene ($CH_2=CH_2$) or propylene ($CH_3-CH=CH_2$) to form a growing ~~&~~ polymer chain.

The stereochemistry of the polymer produced by zieglar-Natta polymerization depends on the type of catalyst used. for ex- titanium based catalysts produce polymers with a high degree of isoactivity, which means that all of the repeating units in the polymer units in the polymer chain are oriented in the same way. Zirconium-based catalysts produce polymers with a high degree of syndiotacticity, which means that the repeating units in the polymer chain are oriented alternately.

Zieglar-Natta polymerization is a very versatile process that can be used to produce a wide variety of polymers with different properties. The polymers produced by zieglar-Natta polymerization are typically linear, high molecular weight & have a higher degree of crystalinity. They are used ~~to~~ in a wide variety of applications including packaging fibers & plastics.

Q.4. Discuss any 5 commercially important polymers.

Ans: (1) Polyethylene (PE):- PE is the most widely produced polymer in the world, with a global production of over 100 million tons per year. It is a versatile & inexpensive polymer that is used in a widely variety of applications including packaging, films, bottles & fibres. PE is also used in a variety of construction applications, such as pipes, siding & insulation.

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② Polypropylene (PP):- PP is the second most widely produced polymer in the world, with a global productn of over 80 million tons per year. It is a strong and durable polymer that is resistant to heat and chemicals. PP is used in a variety of applications, including fibres, carpets, packaging & automotive parts.

③ Polyvinyl chloride (PVC):- PVC is a versatile polymer that is used in a wide variety of applications including pipes, windows, siding & flooring. PVC is also used in a variety of electrical applications such as insulation & wire cable coatings.

④ Polyethylene terephthalate (PET):- PET is a clear & strong polymer that is used in a variety of applications, including bottles, films & fibres. PET is also used in a variety of food packaging applications such as trays and containers.

⑤ Polystyrene (PS): PS is a hard & brittle polymer that is used in a variety of applications, including packaging, toys & appliances. PS is also used in a variety of disposable products such as cups, plates & utensils.